Luis Henrique Jung

Software Engineer · Generative AI & Full Stack

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience, focused on applied Generative AI and production backends in Python (FastAPI) and Java (Spring Boot). Shipped a LangChain AI agent on Google Gemini with a custom MCP (Model Context Protocol) server, multimodal input, and LangSmith observability, saving ~2 minutes per workplace risk assessment. Built a vector-search face recognition platform (OpenCV + Qdrant) with 20,000 enrolled faces that cut PPE handout time from ~3 minutes to ~20 seconds.

CORE COMPETENCIES

Generative AI · AI Agents · LLM Integration · Prompt Engineering · Computer Vision · Vector Search · REST API Design · Microservices · System Architecture · CI/CD & DevOps · Cross-Platform Mobile

WORK EXPERIENCE

RSData 2023 – Present

Full Stack Engineer

- Developed and maintained full-stack applications for **10 client projects** using **Spring Boot**, **Angular**, **NestJS**, **Python (FastAPI)**, and **Flutter**, owning features from REST API design to production deployment
- Engineered a biometric PPE delivery platform that cut handout time from **~3 minutes to ~20 seconds (~89% faster)**, running **600ms face recognition** over **20,000 enrolled faces** with **OpenCV**, **Qdrant vector search**, and **human.js**, plus native **Android** fingerprint authentication for **1,000+ users**
- Built an AI-powered workplace risk management app (**Flutter**) with a **LangChain** agent on **Google Gemini**, a custom **MCP server** exposing full app context to the agent via tool calling, and **LangSmith** observability, saving **~2 minutes per risk assessment** through one-tap recommendations from activity descriptions and workplace images
- Architected **RESTful APIs** and **microservices** with **Spring Boot**, **JPA/Hibernate**, and **SQL Server**, integrating services through **Apache Kafka** event-driven messaging
- Implemented **CI/CD** pipelines with **GitLab CI**, **Jenkins**, and **SonarQube** quality gates, replacing manual deployments with automated builds and code-quality checks
- Co-defined system architecture and the automated testing strategy (**JUnit**, **Mockito**, **AssertJ**), making unit and integration testing part of the continuous delivery flow

OM Tecnologia para Negócios 2022

Junior Software Developer

- Developed **Android (Java)** and **Flutter** apps for client projects, delivering features end to end, from UI implementation to application logic

PROJECTS

Biometric PPE Delivery Platform Production

Fullstack biometric verification system for fast PPE delivery in industrial environments: ~3 min → ~20 s per handout, 600ms face recognition over 20,000 enrolled faces (OpenCV + Qdrant vector search), face-based employee search, and fingerprint authentication on mobile (Android native) and web (Angular + human.js). Core API on Spring Boot + SQL Server, face engine on FastAPI.

Android Native, Angular, Spring Boot, FastAPI, OpenCV, human.js, Qdrant, SQL Server

AI-Powered Risk Management Production

Flutter app for workplace risk management with an integrated AI agent (Google Gemini via LangChain, monitored with LangSmith). Custom MCP server gives the agent full app context; chat UI with contextual action buttons turns activity descriptions and workplace images into one-tap risk recommendations, saving ~2 minutes per assessment. Core API on Spring Boot + SQL Server.

Flutter, Dart, Spring Boot, SQL Server, FastAPI, LangChain, LangSmith, Gemini, MCP

EDUCATION

Postgraduate in Software Engineering for Mobile Devices — UNINTER

BSc in Analysis and Systems Development — UNIVATES

SKILLS

AI/ML: Generative AI, LangChain, LangGraph, LangSmith, MCP, RAG, Prompt Engineering, Tool Calling, Embeddings, Vector Databases (Qdrant), OpenCV, AI Agents

Backend: Java (Spring Boot), Python (FastAPI), Node.js (NestJS), Microservices, Apache Kafka, REST APIs, JPA/Hibernate

Frontend: Angular, TypeScript, JavaScript · **Mobile:** Flutter, Dart, Android (Java)

Databases: PostgreSQL, SQL Server, MySQL, JPQL/HQL

DevOps: Git, Docker, Kubernetes, GitLab CI/CD, Jenkins, SonarQube, Maven

Testing: JUnit, Mockito, AssertJ (unit and integration testing) · **Observability:** Prometheus, Grafana, LangSmith

Languages: English (Intermediate), Portuguese (Native)